DECENT COACHING CLASSES

X Semi

Mind Your Math - IX

Time: 1 Hour 30 Minutes **Maximum Marks: 30**

Note:

(1) All questions are compulsory.
(2) Use of calculator is not allowed.
(3) The numbers on the right side indicate full marks.
(4) In case of MCQs, write the alphabet of the correct answer only.

Q.1 – [8 marks]

Q.1 (A) Four alternative answers are given for each sub-question. Choose the correct one. **[4]**

(1) The value of the following determinant is

$$\begin{vmatrix} 2 & 3 \\ 1 & 4 \end{vmatrix}$$

(a) 5 (b) 11 (c) −5 (d) 10

(2) Which of the following is a quadratic equation?

(a) $x + 3 = 0$ (b) $x^2 + 2x + 1 = 0$
(c) $x^3 = 8$ (d) $2x - 5 = 1$

(3) The common difference of the AP 2, 5, 8, 11, … is

(a) 2 (b) 3 (c) 5 (d) −3

(4) If $\triangle ABC \sim \triangle PQR$, then $AB/PQ = BC/QR =$

(a) AC/PR (b) AC/PQ (c) AB/PR (d) BC/PR

Q.1 (B) Solve the following sub-questions. **[4]**

(1) Write the next term of the AP 3, 6, 9, … [1]

(2) Write the roots of the equation $(x - 2)(x - 3) = 0$. [1]

(3) Find the value of D for the equations $2x + 3y = 5$ and $x + y = 2$. [1]

(4) The sides of two similar triangles are in the ratio 1 : 2. Find the ratio of their areas. [1]

Q.2 (A) and (B) – [12 marks]

Q.2 (A) Complete any TWO of the following activities. **[4]**

(1) Solve the equations $x + y = 5$ and $x - y = 1$ by Cramer's rule. [2]

$D = \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} = (-1) - (1) =$ _____

$Dx = \begin{vmatrix} 5 & 1 \\ 1 & -1 \end{vmatrix} =$ ____ − ____ = ____

$Dy = \begin{vmatrix} 1 & 5 \\ 1 & 1 \end{vmatrix} =$ ____ − ____ = ____

$x = Dx / D =$ _____ $y = Dy / D =$ _____

(2) Find the 10th term of the AP 2, 4, 6, 8, … [2]

$a =$ _____, $d =$ _____, $n = 10$

$t_n = a + (n - 1)d =$ ____ + (10 − 1) × ____ = ____ + ____ = _____

(3) Solve $x^2 - 5x + 6 = 0$ by factorisation. [2]

$x^2 - 5x + 6 = 0$ ∴ $x^2 -$ ____$x -$ ____$x + 6 = 0$

∴ $x(x -$ ____$) - 2(x -$ ____$) = 0$ ∴ $(x -$ ____$)(x -$ ____$) = 0$ ∴ $x =$ ____ or $x =$ ____

Q.2 (B) Solve any FOUR of the following. **[8]**

(1) Solve by elimination method: $x + y = 7$ and $x - y = 3$. [2]

(2) Solve $x^2 - 7x + 12 = 0$. [2]

(3) Find the sum of the first 10 terms of the AP 1, 2, 3, 4, … [2]

(4) In $\triangle ABC$, DE ∥ BC with D on AB and E on AC. If AD = 2, DB = 3 and AE = 4, find EC. [2]

(5) Find the values of the following determinants. [2]

(i) $\begin{vmatrix} 5 & 3 \\ 2 & 4 \end{vmatrix}$ (ii) $\begin{vmatrix} -1 & 2 \\ 3 & 4 \end{vmatrix}$

Q.3 – [6 marks]

Q.3 Solve any TWO of the following. **[6]**

(1) Solve the following equations using Cramer's rule: $2x + y = 5$ and $x + y = 3$. [3]

(2) Solve $x^2 + 6x + 8 = 0$ by factorisation. [3]

(3) The first term of an AP is 5 and the common difference is 3. Find its 8th term and the sum of its first 8 terms. [3]

Q.4 – [4 marks]

Q.4 Solve any ONE of the following. **[4]**

(1) The areas of two similar triangles are 16 cm² and 25 cm². If a side of the smaller triangle is 8 cm, find the corresponding side of the larger triangle. [4]

(2) The sum of two numbers is 12 and their difference is 4. Find the numbers. [4]

— End of Question Paper —